

If blood squirts backward through a defective valve, it may produce an abnormal sound called a **heart murmur**. Some people are born with heart murmurs. In others, the valves may be damaged as a result of infection (for instance, from rheumatic fever, an inflammation caused by infection with certain bacteria). When a valve defect is severe enough to endanger health, surgeons may implant a mechanical replacement valve. However, not all heart murmurs are caused by a defect, and most valve defects do not reduce the efficiency of blood flow enough to warrant surgery.

Maintaining the Heart's Rhythmic Beat

In vertebrates, the heartbeat originates in the heart itself. Some cardiac muscle cells are autorhythmic, meaning they can contract and relax repeatedly without any signal from the nervous system. In fact, these rhythmic contractions continue in tissue removed from the heart and placed in a dish in the laboratory! Given that each of these cells has its own intrinsic contraction rhythm, how are their contractions coordinated in the intact heart? The answer lies in a group of autorhythmic cells located in the wall of the right atrium, near where the superior vena cava enters the heart. This cluster of cells, called the **sinoatrial (SA) node**, acts as a *pacemaker*, setting the rate and timing at which all cardiac muscle cells contract. (In contrast, some arthropods have pacemakers located in the nervous system, outside the heart.)

The SA node produces electrical impulses much like those produced by nerve cells. Because cardiac muscle cells are electrically coupled through gap junctions (see Figure 6.30), impulses from the SA node spread rapidly within heart tissue. These impulses generate currents that can be measured when they reach the skin via body fluids. In an **electrocardiogram (ECG or EKG**, from the German spelling), electrodes placed on the skin record the currents, thus measuring electrical activity of the heart. The graph of current against time has a shape that represents the stages in the cardiac cycle (**Figure 42.8**).

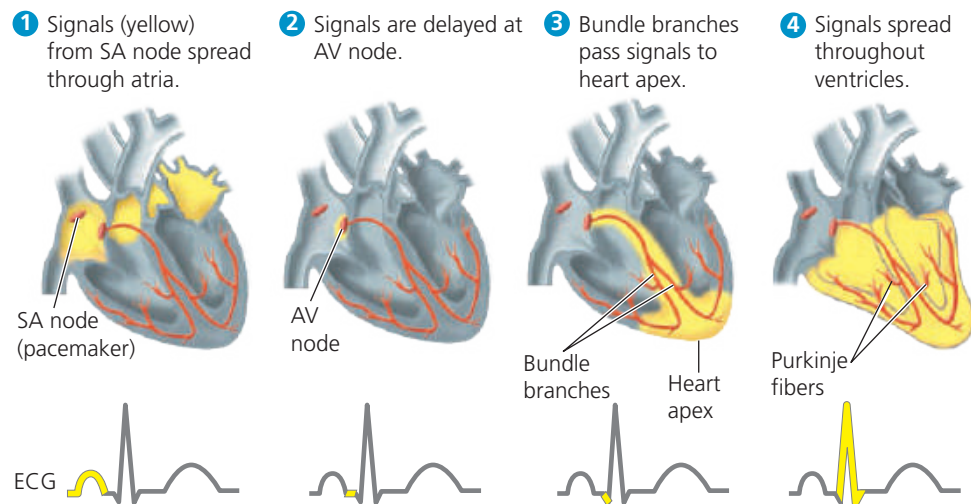
Impulses from the SA node first spread rapidly through the walls of the atria, causing both atria to contract in unison. During atrial contraction, the impulses originating at the SA node reach other autorhythmic cells located in the wall between

the left and right atria. These cells form a relay point called the **atrioventricular (AV) node**. Here the impulses are delayed for about 0.1 second before spreading to the heart apex. This delay allows the atria to empty completely before the ventricles contract. Then the signals from the AV node are conducted to the heart apex and throughout the ventricular walls by specialized structures called bundle branches and Purkinje fibers.

Physiological cues alter heart tempo by regulating the pacemaker function of the SA node. Two portions of the nervous system, the sympathetic and parasympathetic divisions, are largely responsible for this regulation. They function like the accelerator and brake in a car. For example, when you stand up and start walking, the sympathetic division speeds up your pacemaker. The resulting increase in heart rate provides the additional O₂ needed by the muscles that are powering your activity. If you then sit down and relax, the parasympathetic division slows down your pacemaker, decreasing your heart rate and thus conserving energy. Hormones secreted into the blood also influence the pacemaker. For instance, epinephrine, the “fight-or-flight” hormone secreted by the adrenal glands, speeds up the pacemaker. A third type of input that affects the pacemaker is body temperature. An increase of only 1°C raises the heart rate by about 10 beats per minute. This is the reason your heart beats faster when you have a fever.

Having examined the operation of the circulatory pump, we turn in the next section to the forces and structures that influence blood flow in the vessels of each circuit.

▼ **Figure 42.8 The control of heart rhythm.** Electrical signals follow a set path through the heart in establishing the heart rhythm. The diagrams at the top trace the movement of these signals (yellow) during the cardiac cycle; specialized muscle cells involved in controlling of the rhythm are indicated in orange. Under each step, the corresponding portion of an electrocardiogram (ECG) is highlighted (yellow). In step 4, the portion of the ECG to the right of the “spike” represents electrical activity that reprimed the ventricles for the next round of contraction.



WHAT IF? If your doctor gave you a copy of your ECG recording, how could you determine what your heart rate had been during the test?